Journal of Aging & Social Policy

ISSN: 0895-9420 (Print) 1545-0821 (Online) Journal homepage: www.tandfonline.com/journals/wasp20

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To cite this article: Vaneh E. Hovsepian, Douglas M. Sloane, K. Jane Muir & Matthew D. McHugh (2026) Mortality Among the Dementia Population in Not-For-Profit Hospitals with Better Nursing Resources, Journal of Aging & Social Policy, 38:1, 1-15, DOI: 10.1080/08959420.2023.2297596

To link to this article: https://doi.org/10.1080/08959420.2023.2297596

Published online: 31 Jan 2024.

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JOURNAL OF AGING & SOCIAL POLICY
2026, VOL. 38, NO. 1, 1–15
https://doi.org/10.1080/08959420.2023.2297596

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Mortality Among the Dementia Population in Not-For-Profit Hospitals with Better Nursing Resources

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ABSTRACT

The dementia population has higher rates of mortality during hospital stays than those without dementia. The aim of this study is to examine the relationship between ownership status (i.e. for-profit vs. not-for-profit) and nursing resources (i.e. nurse work environment, nurse-to-patient staffing, and nurse education) on 30-day mortality among post-surgical older adults with dementia. A cross-sectional analysis of linked American Hospital Association, Medicare claims, and nurse survey data was conducted using multi-level logistic regression models. We examined these models to assess the relationship between ownership status and 30-day mortality after adjusting patient and hospital characteristics. We also analyzed the relationship between the hospital ownership status and the 30-day mortality, after considering the three nursing resources. Older adults with dementia who received care in hospitals with not-for-profit status were less likely to die within 30 days of admission following surgery compared to those treated in hospitals with for-profit hospital status (i.e. odds ratio 0.82, 95% confidence interval 0.73–0.92, $p = <.001$). In addition, the odds ratios estimating the association between ownership and mortality were similar across the different models of the three nursing resources with and without those controls (i.e. 0.88 vs. 0.83 vs. 0.82). Surgical patients with dementia had better outcomes when cared for in not-for-profit hospitals, particularly with greater levels of nurse education and nurse staffing. The relationship between profit status and mortality was partly explained by the lower levels of nurse staffing and education in for-profit vs. not-for-profit hospitals.

ARTICLE HISTORY
Received 24 May 2023
Accepted 23 October 2023

KEYWORDS
Dementia; mortality; nursing resources; ownership status

Introduction

The number of individuals aged 65 and older in the United States (U.S.) is expected to rise substantially by 2030 (Gaugler et al., 2022). As the number of older adults in the U.S. grows rapidly, the numbers of new and existing cases of Alzheimer's Disease and Related Dementia [(ADRD); hereafter "dementia"]

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are also predicted to increase (Gaugler et al., 2022). In addition, the use of health care services including hospital admission is also soaring (Shepherd et al., 2019). In the U.S., older adults with dementia have more than double the rates of hospital admissions and stays per year compared to the older adults without dementia (Gaugler et al., 2022; Zhu et al., 2015).

In a recent study, 32% of Medicare beneficiaries with dementia had at least one hospitalization compared to 15% of beneficiaries without dementia (Gaugler et al., 2022). Hospitalizations are associated with poor patient outcomes among older adults with dementia, such as increased likelihood of mortality (Möllers et al., 2019; Rao et al., 2016). In a study of the dementia population, researchers discovered that older adults with dementia were at 7% greater risk of death during a hospital stay compared to patients without dementia (Beydoun et al., 2015). The risk of mortality is especially pronounced among older adults with dementia following surgery due to factors such as increased prevalence of delirium and impaired communication (Gaugler et al., 2022; Hu et al., 2012; Tanaka, 2016). As older adults with dementia are more likely to be hospitalized and have higher mortality rates in the hospitals, it is crucial to find ways to minimize poor outcomes among them.

Registered nurses (RNs) account for one of the largest workforces to provide direct care to patients in hospitals. Moreover, RNs oversee and implement tasks that ensure patient safety, which is particularly vital for older adults with dementia as hospitals can be overstimulating due to patient alarms and disorientation which place patients at risk for falls (Digby et al., 2017; Dresser, 2012). Factors such as nursing resources, including nurse education (i.e., proportion of nurses with at least a Bachelor of Science in Nursing [BSN] degree), nurse staffing (i.e., the ratio of patients-to-nurses), and the nurse work environment (i.e., organizational context that facilitates high-quality care delivery), ensure that nurses can address the specific needs of patients during hospitalization. For example, increasing the proportion of BSN educated RNs is linked to lower mortality rates among other populations such as patients wish sepsis (French et al., 2022a; Kumar et al., 2022; White et al., 2018). Additionally, increasing nurse staffing has been associated with lower mortality rates in hospitals (Dierkes et al., 2022; French et al., 2022a; Kumar et al., 2022). Lastly, better nursing environments have also been associated with better patient outcomes such as mortality (Bettencourt et al., 2020; French et al., 2022a). However, little is known about how nursing resources are associated with the mortality rate of older adults with dementia.

Additionally, to our knowledge previous studies have not explored the distribution of nursing resources across hospitals with various ownership statuses (e.g., not-for-profit and for-profit). While over the past decades the prevalence of for-profit status among US hospitals has increased due to reasons such as private equity investments, not-for-profit hospitals still remain a substantial percentage (49%) of the hospitals (Daniel M. Fox, 2015; American Hospital,

2022). Hospitals seek not-for-profit designation to be exempt from taxes, be eligible for tax-exempt bonds and donations, and receive public trust (Daniel M. Fox, 2015; Shen, 2002). However, not-for-profit hospitals are forbidden from sharing profits with employees, and they must use any profits to further their organizational purposes or missions. On the contrary, for-profit hospital owners control their organizations and have the right to profits the hospitals make after all prior obligations have been paid (Furubotn & Pejovich, 1972; Jeurissen et al., 2021). Evidence suggests that for-profit hospitals may be established in areas with well-insured patient populations, while disproportionately offering high-cost, profit driven interventions (such as surgical operations) over less profitable services (e.g., mental health treatment; Horwitz, 2005; Hoxha et al., 2017). Furthermore, previous studies have shown that for-profit hospitals are associated with poorer patient outcomes such as mortality compared to not-for-profit hospitals (Chiu et al., 2020; Shen, 2002). Yet, evidence on how nursing resources are distributed across not-for-profit and for-profit hospitals providing care to post-surgical older adults with dementia is limited.

The purpose of this study is to examine the relationship between ownership status and nursing resources (i.e., nurse work environment, nurse-to-patient staffing, and nurse education) on 30-day mortality among post-surgical older adults with dementia. The findings of this study could potentially inform healthcare policy and practice by identifying strategies to improve the quality of care for older adults with dementia undergoing surgery within the context of changing health care ownership landscapes.

Methods

Study design and data sources

This was a cross-sectional analysis of secondary data from 4 states (California, Florida, New Jersey, and Pennsylvania). We used 2016 data from the following sources: the American Hospital Association (AHA) Annual Survey of Hospitals, Medicare Provider Analysis and Review (MedPAR) files, Beneficiary Annual Summary File (BASF), and RN4CAST-US, a survey of registered nurses (RNs). This study was reviewed by the University of Pennsylvania Institutional Review Board.

Data sources

In addition to providing information about our main predictor variable (i.e., hospital ownership status), the AHA Annual Survey of Hospitals also included hospital characteristics used as covariates in the adjusted regression models (i.e., size, teaching status, and high-technology procedure capabilities). MedPAR included data on inpatient hospitalizations, including the outcome for this study (i.e., 30-day mortality) while the BASF provided Medicare beneficiary demographics for the

patients. Lastly, the RN4CAST-US survey was used to obtain nurse reports of nursing resources including the nurse work environment, nurse staffing, and nurse education.

RN4CAST nurse survey data

Surveys were mailed to the home addresses of approximately 231,000 RNs between January 2015 and December 2016 in the 4 states (California, Florida, New Jersey, and Pennsylvania). Respondents had the option of completing the survey by mail or online. The survey included questions on a range of topics related to nursing practice, nurse demographic information, and the hospitals where nurses were employed. The name of the primary employer was also obtained which allowed the data derived from the surveys to be aggregated to the hospital level and merged with the AHA and patient data. The survey achieved a response rate of 26% ($n = 52{,}510$; Lasater et al., 2019). The survey response rate is similar to many recent surveys, and it yielded an average of 32 nurse responses per hospital, enough to produce reliable estimates of nurse staffing and the nurse work environment (Lasater et al., 2019). This survey has been widely used in previous studies (Kutney-Lee et al., 2021; Lasater & Mchugh, 2016; Porat-Dahlerbruch et al., 2022) and additional details about the methodology are provided elsewhere (Lasater et al., 2019).

Sample

We limited our analysis to patients in non-federal, acute care hospitals that had 10 or more nurse survey respondents – the vast majority of hospitals had more than 20 nurse respondents. While the inclusion of some hospitals with fewer than 20 respondents may produce less reliable hospital-level measures, we wanted a sample of hospitals that represented small ones in addition to large ones. Our patient sample included Medicare fee-for service beneficiaries aged 65 and older who underwent general, orthopedic, or vascular surgery and had been diagnosed with ADRD. Surgical patients were identified using a set of surgical diagnosis-related group (DRG) codes used in prior work (Dierkes et al., 2021; French et al., 2022b; Kutney-Lee et al., 2021). The presence of ADRD was identified using the ADRD flag available in the Chronic Conditions Data Warehouse (CCW). This flag is generated by the Centers for Medicare & Medicaid Services based on an algorithm that searches for designated International Classification of Diseases (ICD-10) for the year of 2016 codes. The CCW flag for ADRD is based on codes that identify Alzheimer's disease, vascular dementia, and frontotemporal dementias, in addition to more generic diagnoses for dementias without a specified underlying disease process. The final sample included 47,115 Medicare beneficiaries cared for in 411 non-federal acute care hospitals across the 4 states.

Measures

Predictor variable. The ownership status variable, which contrasted for-profit and not-for profit hospitals, was obtained from the AHA Annual Survey. Based on these data, for-profit hospitals were owned by investors such as individuals, partnerships, and corporations. Hospitals that were owned by the state, county, city, city-county, hospital district or authority, church operated, and other not-for profit organizations were categorized as not-for profit.

Outcome variable. Patient outcomes were obtained from the MedPAR database and included an indicator of whether patients died in the hospital within 30 days of admission. For Medicare beneficiaries who had more than one admission in one year, we used the first hospitalization for our analysis.

Covariates. Several hospital and patient characteristics were also included in the analysis to account for potential confounding. These included hospital characteristics obtained from the AHA Annual Survey, including size (i.e., ≤100 beds, 101–250 beds, or > 250 beds), teaching status (i.e., nonteaching, minor teaching, or major teaching), and technology capabilities (i.e., ability to perform open heart surgeries and/or major organ transplants) as well as patient characteristics obtained from the MedPAR and BASF databases, including age, sex, surgical type based on the DRG, and 27 comorbidities originally defined by Elixhauser (Elixhauser et al., 2010).

The nursing resources (i.e., nurse work environment, nurse-to-patient staffing, and nurse education) were constructed using the 2016 RN4CAST-US survey and have been detailed in previous studies (Aiken et al., 2009; 2014; Lasater et al., 2021; Sloane et al., 2018). The nurse work environment was measured using the 31-item Practice Environment Scale of the Nursing Work Index (PES-NWI). The PES-NWI subscale is constructed from questions asked of nurses related to nurse-physician relations, staffing and resource adequacy, and nursing leadership. At the hospital level, work environment was measured by averaging nurses' reports of the work environment across all nurses within each hospital, and ultimately scoring each hospital on a four-point scale from poor to excellent. The nurse-to-patient staffing ratio for each nurse was measured by asking nurses how many patients were assigned to them on their last shift. Consistent with previous research (French et al., 2022b; Sloane et al., 2018), nurse-to-patient staffing was measured as the hospital's average patient-to-nurse ratio. BSN education was a continuous variable indicating percentage of all staff nurses per hospital holding at least a BSN degree. We used these hospital-level measures to account for the fact that individuals undergoing surgery are usually cared for on multiple units during their hospitalizations.

Data analysis

Characteristics and outcomes of patients in the sample were examined using descriptive statistics. To determine whether there were significant differences between the patients in for-profit and not-for profit hospitals, and differences in the other characteristics (i.e., size, teaching status, etc.), we used chi-square tests except for age where we used a simple t-test. For hospital size, teaching status, and state variables where we used Fisher's exact test because we had fewer than five observations for some of the categories for these variables. We used multivariate and multi-level generalized estimating equations (logistic regression models) to take account of potential confounding factors and to account for the clustering of patients within hospitals. We first (Model 1) estimated and examined the coefficients of a model that allowed an association of 30-day mortality with hospital ownership among older adults with dementia, controlling for patient characteristics and hospital characteristics. Then, we studied the relationship between the nursing resources and 30-day mortality among this population (Model 2). Finally, we assessed the relationship between hospital ownership status and 30-day mortality, after taking account of the nursing resources, among the dementia population (Model 3). All regression models were adjusted for both hospital characteristics (i.e., hospital size, teaching status, and technology status) and patient characteristics (i.e., age, sex, DRGs, and comorbidities) characteristics. The reference group for ownership status was for-profit hospitals. Nursing resources were measured as ordered and linear variables. SAS 9.4 was used to conduct the analysis.

Results

Patient and hospital characteristics

The characteristics, including mortality, of older adults with dementia who underwent surgery ($n = 47{,}115$) in for-profit and not-for profit hospitals are shown in Table 1. Patient characteristics such as age, sex, race, and surgical procedure for patients in for profit and not-for profit hospitals were similar, in spite of the fact that some differences are statistically significant because of the large size of the patient sample. For instance, the mean age of patients in our sample was approximately 83 years old across both groups. Most of the patients were female (63% to 64%) and belonged to the non-Hispanic white race/ethnicity group (87% to 88%) in both groups. Overall, most patients were admitted for orthopedic (64%) surgery, followed by general (24%) then vascular (13%) surgeries, and most of the older adults with dementia in our sample, in both for-profit and not-for profit hospitals, had high rates of comorbidities such as uncomplicated hypertension (58% overall), hypothyroidism (23% overall), and chronic pulmonary disease (21% overall). Compared to for-profit hospital patients, not-for profit patients had

significantly lower rates of diabetes without chronic complications, and higher rates of diabetes with chronic complications, peripheral vascular disease, and moderate renal failure. Patients receiving care from for-profit hospitals had higher rates of 30-day mortality compared to patients receiving care from not-for profit hospitals (7% vs. 6%; $p = .002$).

Characteristics of the study hospitals overall and by hospital ownership status are described in Table 2. Most of (57%) not-for profit hospitals were large (>250 bed) compared to hospitals in the for-profit group (43%; $p = .053$), and more of them (48% vs. 19%) were non-teaching hospitals. Most of for-profit hospitals (80%) had minor teaching status compared to not-for profit hospitals (40%; $p = <.001$). Technology status was similar across both groups. Of the four states, California had the highest percentage of hospitals classified as not-for profit (43%) and Florida had the highest percentage of for-profit hospitals (62%).

Table 1. Characteristics and outcomes of older adults with dementia across for-profit and not-for profit hospitals.

Patient Characteristics	Total (*n* = 47,115)	For-profit (*n* = 89,32)	Not-for profit (*n* = 38,183)	*P*
Age, mean (SD)	82.75 (8.04)	82.84 (7.96)	82.73 (8.06)	0.243
Sex n (%)				0.416
Male	17351 (36.8)	3256 (36.5)	14095 (36.9)	
Female	29764 (63.2)	5676 (63.6)	24088 (63.1)	
Race/ethnicity, n (%)				0.009
Non-Hispanic White	40974 (86.6)	7845 (87.7)	33139 (86.8)	
Non-Hispanic Black	2594 (5.5)	479 (5.4)	2115 (5.5)	
Hispanic	1534 (3.3)	276 (3.1)	1258 (3.3)	
Asian	1023 (2.2)	196 (2.2)	827 (2.2)	
Other (i.e., North American Native, unknown, and other)	990 (2.1)	146 (1.6)	844 (2.2)	
Elixhauser comorbidities				
Deficiency anemias	9848 (20.9)	1973 (22.1)	7875 (2.6)	0.002
Cerebrovasculardisease	2505 (5.3)	408 (4.6)	2097 (5.5)	<0.001
Heart failure	8107 (17.2)	1484 (16.6)	6623 (17.4)	0.099
Depression	8200 (17.4)	1539 (17.2)	6661 (17.4)	0.630
Diabetes without chronic complications	8137 (17.3)	1690 (18.9)	6447 (16.9)	<0.001
Diabetes with chronic complications	5180 (10.9)	858 (9.6)	4322 (11.3)	<0.001
Hypertension, complicated	10030 (21.3)	1836 (2.6)	81194 (21.5)	0.060
Hypertension, uncomplicated	27150 (57.6)	5254 (58.8)	21896 (57.3)	0.011
Chronic pulmonary disease	10058 (21.4)	1982 (22.2)	8076 (21.6)	0.031
Obesity	3805 (8.1)	708 (7.9)	3097 (8.1)	0.565
Peripheral vascular disease	5452 (11.6)	972 (1.9)	4480 (11.7)	0.024
Renal failure, moderate	7502 (15.9)	1353 (15.2)	6149 (16.1)	0.026
Renal failure, severe	1473 (3.1)	264 (2.9)	1214 (3.2)	0.274
Hypothyroidism	10820 (22.9)	2019 (22.6)	8801 (23.1)	0.368
Surgical procedure, n (%)				<0.001
General	11052 (23.5)	1991 (22.3)	9061 (23.7)	
Orthopedic	30101 (63.9)	5721 (64.1)	24380 (63.9)	
Vascular	5962 (12.7)	1220 (13.7)	4742 (12.4)	
Patient outcome, n (%)				
30-day mortality	2969 (6.3)	626 (7.0)	2343 (6.1)	0.002

Table 2. Characteristics of study hospitals.

Hospital Characteristics	Overall (*n* = 411)	For-profit (*n* = 79)	Not-for profit (*n* = 332)	*P*
Hospital size, n (%)				0.053
⇐100 beds	14 (3.4)	2 (2.5)	12 (3.6)	
101–250 beds	174 (42.3)	43 (54.4)	131 (39.5)	
>250 beds	223 (54.3)	34 (43.0)	189 (56.9)	
Teaching status, n (%)				<.001
Nonteaching	175 (42.6)	15 (18.9)	160 (48.2)	
Minor teaching	196 (47.7)	63 (79.8)	133 (4.1)	
Major teaching	40 (9.7)	1 (1.3)	39 (11.8)	
Technology status, n (%)				0.222
Low	155 (38.1)	25 (32.1)	130 (39.5)	
High (performs open heart surgery and/or organ transplant)	252 (61.9)	53 (67.9)	199 (6.5)	
State, n (%)				<.001
CA	159 (38.7)	17 (21.5)	142 (42.8)	
FL	120 (29.2)	49 (62.0)	71 (21.4)	
NJ	43 (10.5)	1 (1.27)	42 (12.7)	
PA	89 (21.7)	12 (15.2)	77 (23.2)	
Nursing resources				
Nurse staffing (patients per nurse), mean (SD)	4.50 (0.98)	4.72 (1.05)	4.44 (.96)	0.026
Nurse work environment, mean (SD)	2.83 (0.25)	2.67 (.22)	2.87 (.25)	<.001
Percentage BSN nurses, mean (SD)	54% (0.17)	46% (.16)	56% (.16)	<.001

Abbreviations: SD: Standard deviation, CA: California, FL: Florida, NJ: New Jersey, PA: Pennsylvania.

Associations of hospital ownership status on surgical patient mortality

Table 3 displays results from our regression models examining the relationships between hospital ownership status and nursing resources on 30-day mortality of older adults with dementia. In Model 1, we find that, even after controlling for patient and hospital characteristics, older adults with dementia who received care in hospitals with not-for profit status were less likely to die within 30 days of admission following surgery compared to those treated in hospitals with for-profit hospital status, by a factor of 0.82 (i.e., odds ratio [OR] 0.82, 95% confidence interval [CI] 0.73–0.92, $p = <.001$). Models 2a, 2b, and 2c – which estimate the effects of the three nursing resource variables one at a time, after including the same patient and hospital controls, show that there are associations between mortality and the work environment (OR 0.81, 95% CI 0.66–1.00, $p = .045$), RN education level (OR 0.92, 95% CI 0.90–0.96, $p = <.0001$), and the patient-to-nurse staffing ratio (OR 1.07, 95% CI 1.02–1.13, $p = .011$). Lastly, in models 3a, 3b, and 3c, we examined the relationship between hospital ownership status and the 30-day mortality of older adults with dementia, after taking account (in separate models) of the three nursing resources. Model 3a shows no effect of the work environment on mortality, and thus little change in the ownership effect (from 0.82 to 0.84) when we include it in the model. Model 3b shows that each 10% increase in the proportion of BSN-educated nurses was associated with lower odds of 30-day mortality (OR 0.93, 95% CI 0.90–0.96, $p = <.0001$), and that controlling for that

Table 3. Association between hospital not-for profit status and nursing resources and mortality.

	Model 1		Models 2a, 2b, and 2c		Models 3a, 3b, and 3c	
Independent Variables	*OR*	*P*	*OR*	*P*	*OR*	*P*
Ownership status	0.82 (0.73–0.92)	<.001	___	___	0.84 (0.73–0.95)	0.008
(a) Work environment (Continuous)	___	___	0.81 (0.66–1.00)	0.045	0.88 (0.70–1.09)	0.245
Ownership status	___	___	___	___	0.88 (0.78–0.99)	0.029
(b) RN education level	___	___	0.92 (0.90–0.96)	<.0001	0.93 (0.90–0.96)	<.0001
Ownership status	___	___	___	___	0.83 (0.74–0.94)	0.003
(c) Staffing ratio	___	___	1.07 (1.02–1.13)	0.011	1.06 (1.00–1.12)	0.035

All models are adjusted for hospital and patient characteristics including demographics and comorbidities.

association does attenuate somewhat the difference between not-for-profit hospitals and for-profit hospitals (i.e., the OR changes from 0.82 to 0.88, the latter of which is closer to 1.0, which would indicate no difference). Finally, model 3c shows that while each additional patient-per-nurse was associated with significantly higher odds of 30-day mortality (OR 1.06, 95% CI 1.00–1.12, $p = .035$), controlling for that association has no effect on the difference between not-for-profit hospitals and for-profit hospitals (OR changes from 0.82 to 0.83).

Discussion

This is the first study to explore how nursing resources are distributed across hospitals with different ownership status and how the resources as well as ownership status are associated with 30-day mortality among postsurgical older adults with dementia. We found significant differences in mortality rates and nursing resources across not-for-profit and for-profit hospitals. Postsurgical older adults with dementia receiving care from not-for-profit hospitals had lower 30-day mortality rates and received care from nurses with better nursing resources compared to for-profit hospitals. Our findings are consistent with prior research identifying higher mortality rates in for-profit hospitals compared to not-for-profit hospitals (Chiu et al., 2020; Shen, 2002). While profit status and nursing resources had independent direct effects on mortality, the differences in nursing resources accounted for a small component of the advantage patients had in avoiding death in not-for-profit vs. for-profit hospitals. High nurse-to-patient ratios and higher nurse education level are associated with better patient care and lower mortality rates. Policymakers can establish regulations on nurse-to-patient ratios and improve nurse education level to ensure safe staffing levels in

hospitals. Financial incentives, such as reimbursement adjustments based on staffing and education, can encourage hospitals to maintain appropriate nurse staffing and education levels.

There may be several reasons for our finding that ownership is associated with mortality among the dementia population. First, older adults with dementia experience deficits in memory, language, and judgment which translates into difficulties reporting symptoms, adhering to medications, reporting medication side effects, and complying with treatment and follow-up recommendations (Gual et al., 2018; Nelis et al., 2019). Thus, in addition to the studied nursing resources, the dementia population may need more resources for optimal outcomes. For instance, various hospital leadership styles have been shown to influence mortality rates among the dementia population (Cummings et al., 2010). Hospital leadership styles that prioritize patient-centered care, staff well-being, and collaboration tend to have better outcomes for dementia patients. In addition, it is possible that ownership status may be associated with other hospital-level factors that influence mortality rates among older adults with dementia who undergo surgery. For example, hospitals that are not-for-profit may have different funding structures, management practices, or quality improvement initiatives compared to for-profit hospitals, which could translate into patient-centered care efforts and expenditures that are themselves associated with patient outcomes. Therefore, future research should focus on how leadership styles, funding structures, management practices, or quality improvement initiatives are varied across hospitals with different ownership status and how those variations influence mortality among the dementia population. In addition, future studies should also explore how other important covariates (i.e., dual-eligible status of patients, hospital's health system affiliation, safety-net status, and EHR implementation) can impact 30-day mortality among post-surgical older adults with dementia.

Future studies should also focus on the stages of dementia, as each stage requires special care requirements and attention in hospitals, particularly nursing care. Because dementia has mild, moderate, and severe stages (Alzheimer's Association, 2020), older adults with dementia in different stages have different care requirements. Older adults with dementia with moderate and severe dementia have more complex medical needs than those with mild dementia. They face more difficulties communicating and performing routine tasks, including activities of daily living and managing their comorbidities (Alzheimer's Association, 2020), which puts them at a higher risk of mortality. Therefore, future studies should look at how other hospital and patient level factors influence the association between ownership and mortality.

Limitations

This study has some limitations. For example, the study has a cross-sectional design which restricts our ability to assess causality and to make causal claims. Due to our sampling of hospitals, nurses, and patients in 4 US states, the generalizability of our study results may be limited and may be only applicable to the 4 states. In addition, other factors might have impacted the outcome of this study. More specifically, ADRD which is a progressive neurodegenerative disease, has different severity stages that we were not able to assess because Medicare claims data do not have this information. Thus, we were not able to evaluate how each stage is associated with our findings. Lastly, due to challenges involved with the diagnosis of the disease, some older adults with dementia might not have been identified in the Medicare data.

Conclusion

Findings from this study generate important evidence about the association between ownership and mortality among older adults with dementia. We found that profit status, nurse education, and nurse staffing levels were significantly associated with mortality. Taking account of the nursing features had only a small effect on the association between ownership and mortality. Further research is needed to identify the specific factors that contribute to the association between ownership status and mortality rates among older adults with dementia who undergo surgery. This can help inform interventions and policies that aim to improve the quality of care and outcomes for this vulnerable patient population.

Key Points

- Older adults with dementia have higher rates of mortality during their hospital stay compared to those without dementia. Therefore, it is vital to study the potential factors that influence mortality rates in hospitals among this population.
- Surgical patients with dementia had better outcomes when cared for in not-for-profit hospitals.
- The association between profit status and mortality was partly explained by the lower levels of nurse staffing and education in for-profit vs. not-for-profit hospitals.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This work was supported by the National Institute of Nursing Research grant numbers T32 -NR007104 (Aiken, PI) and R01- NR014855 (Aiken/McHugh, PIs).

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Authors' contributions

VH contributed to study conceptualization, writing, data analysis, and interpretation. MM contributed to design and interpretation of data and analysis. MM, DMS, and KJM provided statistical advice and oversight on manuscript development and revisions.

Impact statement

We certify that this work is novel. This is the first study to explore how nursing resources are distributed across hospitals with different ownership status and how they are associated with 30-day mortality among postsurgical older adults with dementia.

Sponsor's role

Sponsors were not involved with the design, methods, subject recruitment, data collection, analysis, or preparation of this manuscript.

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